

# Theo Obadiah Teguh

theo.obadiah@gmail.com | [Website](#) | [GitHub](#) | [LinkedIn](#) | +852 6595 4989

## EDUCATION

### The University of Hong Kong

Sep 2025 - Aug 2026

*Master of Statistics, Focus in Financial Statistics and Data Analytics*

*Hong Kong*

- **CGPA** — 3.77/4.30 (Distinction), **Awards** — Belt and Road Scholarship (Full Tuition, 2025-2026)
- **Coursework** — Quantitative Trading, Time Series, Portfolio Theory, Financial Monte Carlo, Statistical Modelling

### The University of Hong Kong

Sep 2021 - May 2025

*Bachelor of Economics and Finance, 2nd Major in Computer Science*

*Hong Kong*

- **CGPA** — 3.65/4.30 (First Class Honours), **Awards** — Dean's Honours List (2021-2025), HKU Entrance Scholarship (Full Tuition, 2021-2025), Chan Tat Chee Memorial Education Abroad Scholarship (2023-2024)
- **Coursework** — Software Engineering, Data Structures & Algorithms, Operating Systems & Networks, Relational Databases, Object-Oriented & Functional Programming, Mathematical Finance & Econometrics, Statistical Learning

## PROFESSIONAL EXPERIENCE

### Software Engineer Trainee, EQD ([more details](#))

Aug 2026 - Present

*BNP Paribas Corporate and Institutional Banking*

*Hong Kong*

- Designed **market-implied volatility** monitoring and ETL pipelines serving EQD trading and structuring desks.
- System administrator of the internal **pricing platform for exotics**, utilized by APAC clients and internal users.

### Software Engineer Intern ([more details](#))

Dec 2025 - July 2026

*ActusRayPartners, LTD*

*Hong Kong*

- Developed a cross-SMA tracking error monitoring tool with **VBA**, based on **Lagrangian optimization**, adopted firm-wide by the quantitative trading team across different markets.
- Spearheaded the end-to-end automation of daily pricing **factor beta calculations**, replacing Excel workflows with a 5-second execution with **Bloomberg Query Language**.
- Designed an **Outlook** viewing system for market data **ETL pipeline jobs** in the APAC and EU regions, reducing developer team manual effort by 5+ hours weekly.
- Researched **Monte Carlo** for corporate actions and **PCA** or **relevance-weighted liquidity** for regime switching..

### Deep Learning Research Intern ([more details](#))

Jul 2025 - Aug 2025

*Shanghai Jiao Tong University*

*Shanghai*

- Implemented training scripts for **CNNs** with 19M parameters, achieving 93% accuracy on CIFAR-10.
- Organized deep learning dependencies such as **PyTorch** and **Lightning** with **Docker** and **Poetry**.

### Data Science Intern ([more details](#))

Jul 2024 - Dec 2024

*Oxbridge Economics, LLC*

*Hong Kong*

- Designed and implemented a stratified sampling algorithm with **scikit-learn** and **Pandas** with a 56 GB geospatial data warehouse and **Landsat satellite data**, increasing R-squared by 59%.
- Conducted **feature selection** and trained robust **tree-based models** (XGBoost) with regularization, cross-validation, and Bayesian hyperparameter tuning to estimate geospatial variability of soil carbon.

### DevOps Engineer (Research Assistant) ([more details](#))

Mar 2024 - Apr 2024

*Vancouver School of Economics*

*Vancouver*

- Reduced **software deployment** time by 50% via recursive synchronization of dependencies.
- Automated **CI/CD** with **Bash** and **GitLab CI**, mitigating the risk of error in a project of 30+ developers.

### Data Engineer (Research Assistant) ([more details](#))

Jun 2023 - Sep 2023

*HKU Business School*

*Hong Kong*

- Developed an **ETL pipeline** with **Python** and **AWS RDS MySQL** for a 35 GB **XML Morningstar** dataset.
- Reduced end-to-end data ingestion time from ~10 hours to under 1 hour by optimizing pipeline architecture, achieving sustained throughput of 104 Mbps (13x speedup).

## RELEVANT PROJECTS

### Tower Research Capital - Supervised Alpha Research Project ([more details](#))

Dec 2024

- Deployed infrastructure on HKU HPC GPU farm to process 3.4 million Glassdoor records, utilizing BERT-based sentiment embeddings to generate alpha signals, and backtested market-neutral strategies.

### Citadel Trading Challenge - 1st Place ([more details](#))

Oct 2024

- Brainstormed and solved a collaborative mathematical trading simulation game, achieving 1st place amongst 30 select participants across 6 teams from the University of Hong Kong.

## TECHNICAL SKILLS

- **Programming Languages** — Python, R, VBA, C, C++, Java, Haskell, Bash, SQL
- **Statistics and Machine Learning** — Pandas, NumPy, SciPy, scikit-learn, PyTorch, NLTK, ElementTree
- **Cloud and DevOps** — AWS (EC2, S3, EKS, RDS, CloudFront, Route 53), Docker, Karpenter, Git, CI/CD